

Reflective Journal — Module 04

Module 04 Reflective Journal — Exploratory Data Analysis

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Before this lab, I thought data analysis mostly meant looking at numbers and calculating averages. This lab helped me understand that exploratory data analysis (EDA) is an important step before building a machine-learning model. EDA helps us learn what the data contains, notice missing values, and look for patterns that might affect a model's results.

The Titanic dataset made the ideas easier to understand because it included passenger information such as age, gender, passenger class, fare, and survival. The notebook uses summary statistics and several plots to compare these details. A count plot can show how many passengers survived or died. Comparing survival by class and gender helps reveal differences between groups. A histogram helps us look at age distributions, while a box plot can compare ticket fares for passengers who survived and those who did not.

One important thing I learned is that a chart and a summary number answer different questions. An average age gives one useful number, but it does not show the full distribution or whether different groups have different patterns. A plot can make differences, outliers, and skewed data easier to notice. Using both statistics and visualizations gives a better understanding than relying on only one method.

The experimentation section asks us to compare survival by port of embarkation and compare ticket fares by survival outcome. These questions helped me practice choosing a visualization based on the type of data. A count plot is useful for categories such as embarkation port, while a box plot is useful for comparing a numerical variable such as fare across two outcome groups. I also included survival proportions by port because the number of passengers from each port is not the same. Proportions can make group comparisons fairer than raw counts alone.

This lab also reminded me that a relationship does not prove cause and effect. If passengers with higher fares or certain passenger classes had better survival outcomes, fare itself may not be the only reason. Fare can be connected to passenger class and other circumstances. In machine learning, it is important to question what the data actually supports instead of making a conclusion too quickly.

The workflow makes more sense to me now: first load the data, then inspect its structure and missing values, summarize it, visualize important variables, and use the findings to guide later modeling decisions. EDA does not guarantee that a model will be accurate, but it can help identify data quality issues and useful features before modeling begins. My next step is to run all notebook cells in Google Colab, check the actual outputs, and make sure my written interpretations match the charts and values produced.